XIAMEN UNIVERSITY MALAYSIA

Online Quiz 4

Course Code:	COM108		
Course Name:	Engineering Maths (I)		
Question Paper Setter:	Liu Mei Feng		
Academic Session:	2021/09	Question Paper:	A ☐ B ☐
Total No. of Pages:	1	Time Allocated:	1 hours
Additional Materials:	-		
Apparatus Allowed:	-		

INSTRUCTIONS TO CANDIDATES

1. There are THREE questions in this paper. Please answer ALL questions.
2. Please follow the requirement of each question and give the corresponding answers in PDF format.
3. Please note that presentation of solutions (clarity, coherence, conciseness and completeness) is important. Show your work and organize your solution. Answer without proper justification may receive less, or even zero credit.
4. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (15 marks)

Evaluate the definite integral.

(a) [5 marks]

$$\int_{-\infty}^{1} xe^{x}\,dx$$

(b) [5 marks]

$$\int_{6}^{8} \frac{4}{(x-6)^{3}}\,dx$$

(c) [5 marks]

$$\int_{-\infty}^{\infty} xe^{-x^{2}}\,dx$$

Question 2 (15 marks)

Use the partial fraction decomposition to find the integral.

(a) [5 marks]

$$\int \frac{1}{x^2 - 9} dx$$

(b) [5 marks]

$$\int_0^2 \frac{3}{4x^2 + 5x + 1} dx$$

(c) [5 marks]

$$\int \frac{\sin x}{\cos x \, (\cos x - 1)} dx$$

Question 3 (20 marks)

Let *S* be the set given by

$$S = \left\{(x, y) \middle| x \geq 0, 0 \leq y \leq \frac{1}{e^x} \right\}$$

(a) Sketch the graph in the *xy*-plane and indicate the region of the set S. [5 marks]

(b) Calculate the area of the region if it is finite. [5 marks]

(c) Evaluate the arc length of the curve $y = e^{-x}$ between $x = 0$ and $x = 1$. [5 marks]

(d) Evaluate the surface area of the solid generated by revolving S about the x-axis. [5 marks]

- END OF PAPER -